



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 3

6031/3

JUNE 2019 SESSION

2 hours 30 minutes

Additional materials:

Answer paper

Data Booklet

Mathematical tables and/or electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 12 printed pages.

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Section A

Answer any two questions from this section.

- 1 (a) (i) State and explain the trend in the 1st ionisation energies of elements in the same group.
- (ii) Successive ionisation energies of three elements, A, B and C, are given in Table 1.1.

Table 1.1

element	ionisation energies/kJmol ⁻¹			
	1 st	2 nd	3 rd	4 th
A	494	4 560	6 940	9 540
B	418	3 070	4 600	5 860
C	403	2 633	3 860	5 080

State and explain the Group of the Periodic Table to which the three elements are most likely to belong.

[4]

- (b) (i) Explain why the first ionisation energy of Boron is lower than that of Beryllium.
- (ii) Write an equation that represents the second ionisation energy of Boron.

[3]

- (c) (i) Draw an energy level diagram to show the electronic configuration of Boron.

- (ii) Describe the shapes of the s and the p orbitals.

[3]

- (d) The solubility product of iron (II) hydroxide is $x \text{ mol}^3 \text{dm}^{-9}$.

- (i) Calculate the solubility of iron (II) hydroxide in terms of x .

(ii) Describe and explain the effect on the value in (i) of

1. increasing the temperature,
2. dissolving the iron (II) hydroxide in $\text{NaOH}_{(\text{aq})}$.

[5]
[Total: 15]

2 (a) A hydrogen-oxygen fuel cell is shown in Fig. 2.1.

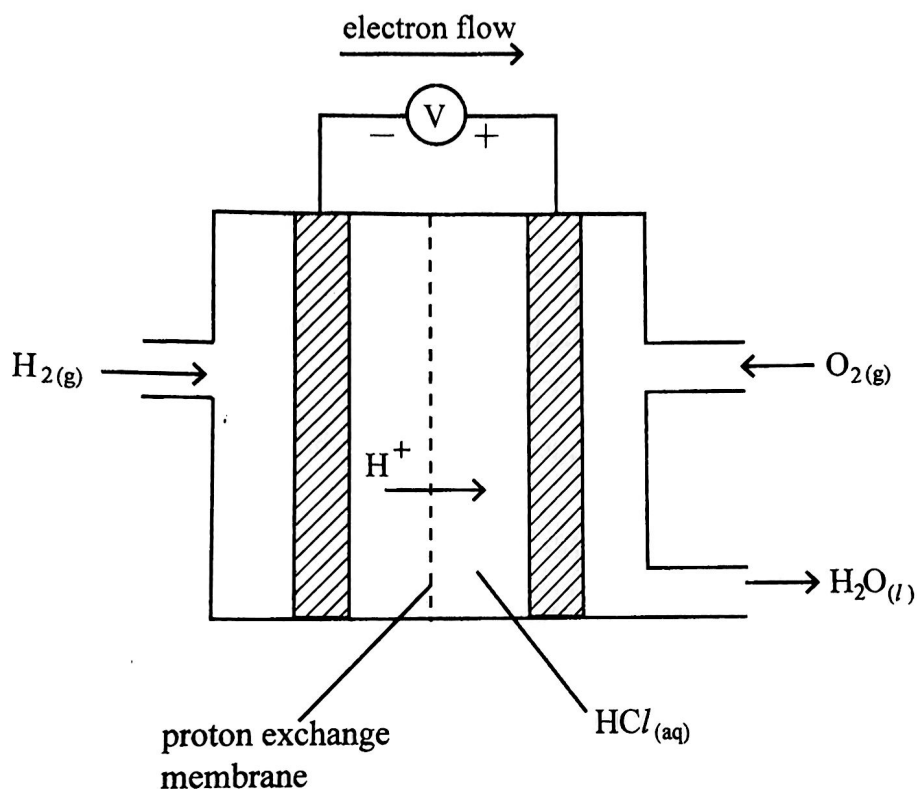


Fig. 2.1

(i) Write the equation for the overall cell reaction occurring in the hydrogen-oxygen fuel cell.

(ii) Calculate E^θ of the fuel cell.

[3]

(b) State any two

- (i) advantages of using fuel cells compared to fossil fuels,
- (ii) limitations to the use of fuel cells.

[3]

- (c) An electrolytic cell was set up to determine the value of the Avogadro constant as shown in Fig. 2.2. 1.90 A of current were passed through the aqueous copper (II) sulphate for 50 minutes.

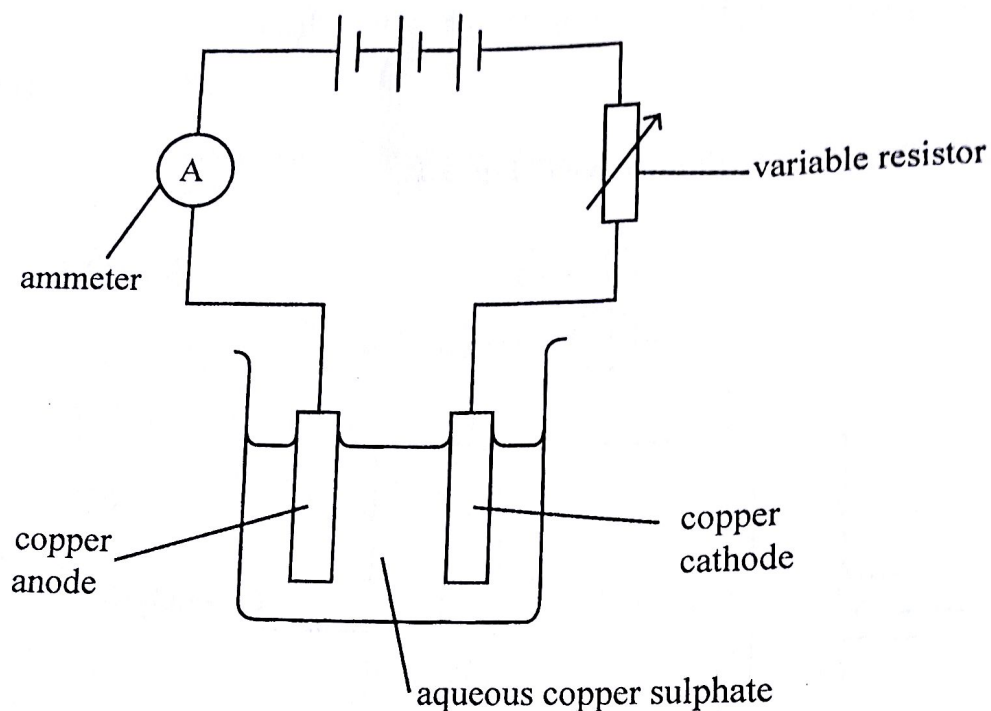


Fig. 2.2

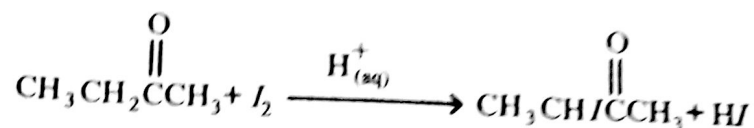
- (i) State any **three** observable changes that occurred.
- (ii) The mass of the cathode changed from 70.14 g to 71.97 g.
Calculate the Avogadro constant.
- (iii) Explain why measuring the decrease in mass of the anode would be preferred to measuring the increase in mass of the cathode.

[9]

[Total 15]

3

- (a) Butanone reacts with iodine in an acid catalysed reaction as shown.



- (i) Describe how the rate of this reaction can be determined.
- (ii) An experimental investigation showed that the reaction is zero order with respect to iodine and that butanone exists in two interconvertible forms shown, in Fig. 3.1.

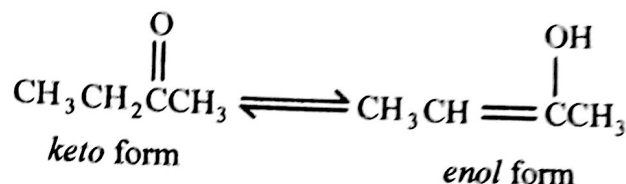


Fig. 3.1

Propose a two step mechanism for the reaction.

[6]

- (b) (i) Sketch graphs, on the same axes, to show how the rate of a reaction varies with concentration of a reactant for a

1. zero order,
2. first order and
3. second order reaction.

- (ii) The overall order of a reaction was found to be third order.

Deduce the units of the rate constant for the reaction.

[4]

- (c) Explain the following observations.

- (i) Some collisions of particles do not lead to chemical reactions.
- (ii) An increase in temperature increases the rate of a reaction.

[5]

[Total:15]

Section B

Answer any **one** question from this section.

- 4 (a) Table 4.1 shows the boiling points of some of the Group (IV) tetrachlorides.

Table 4.1

tetrachloride	boiling point/ $^{\circ}\text{C}$
SiCl_4	58
GeCl_4	87
SnCl_4	114

- (i) State the shape of the tetrachlorides formed by Group (IV) elements.
- (ii) Explain the variation in the boiling points of the chlorides in Table 4.1.
- (iii) Explain why CCl_4 does **not** hydrolyse.
- (iv) Write an equation for the reaction between SiCl_4 and water.

[5]

- (b) Write balanced chemical equations for the following reactions:

- (i) CO and Fe_2O_3
- (ii) SnO_2 and concentrated HCl
- (iii) PbO_2 and concentrated HCl

[6]

- (c) State and explain the acid-base nature of

- (i) CO ,
- (ii) GeO_2 .

[4]

[Total 15]

- 5 (a) **Table 5.1** shows the melting points of the chlorides of the elements sodium to phosphorus.

7

Table 5.1

compound	melting point /°C
sodium chloride	808
magnesium chloride	714
aluminium chloride	sublimes at 178°C
silicon tetrachloride	-70
phosphorus (v) chloride	162

- (i) Explain the variation in the melting points of the chlorides.
- (ii) Write balanced chemical equations for the reaction of each of the following chlorides with water:

1. sodium chloride
2. aluminium chloride
2. phosphorus (V) chloride

[7]

- (b) A 6.6125 g sample of a nitrate of a Group (II) metal, **M**, decomposed on heating to liberate 1 875 cm³ of gas at room temperature and pressure.

- (i) Write an equation for the decomposition of the nitrate.
- (ii) Deduce, with reasons, the identity of **M**.

[5]

- (c) (i) Describe **two** observations that occur when sodium reacts with chlorine.
- (ii) Explain why sodium chloride is used as a food preservative.

[3]

[Total:15]

Section C

Answer any *two* questions from this section

- 6 Fig. 6.1 shows how an organic compound can be converted to compounds **G**, **H** and **I**.

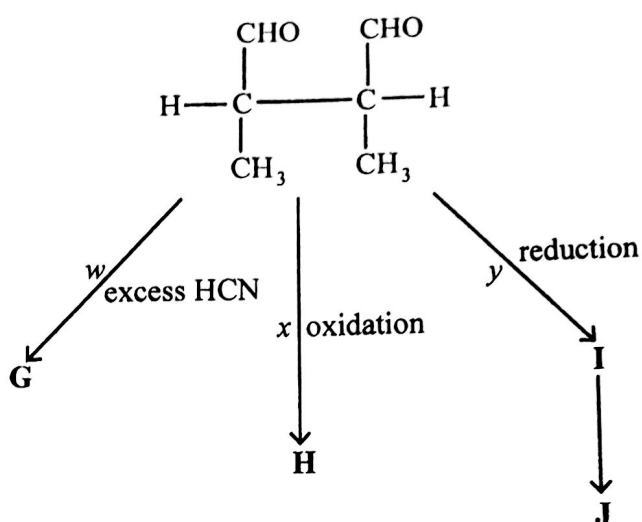


Fig. 6.1

- (a) (i) State, with reasons, the condition(s) for reaction *w*.
 (ii) Name reaction *w*.
 (iii) Describe the mechanism for reaction *w*. [6]
- (b) (i) State the reagents and conditions for
 1. reaction *x*,
 2. reaction *y*.
 (ii) Draw the structures of **H** and **I**.
 (iii) Name compounds **H** and **I**.
 (iv) Products **H** and **I** can react to form compound **J**.
 Name the type of reaction. [7]

(c) Describe a one-step laboratory test to distinguish between compounds **H** and **I**. [2]
[Total 15]

7 (a) **Fig. 7.1** shows the structure of an amino acid, proline.

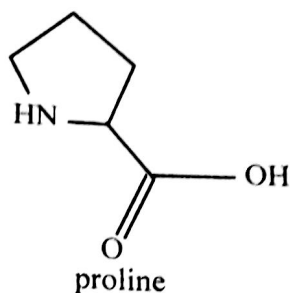
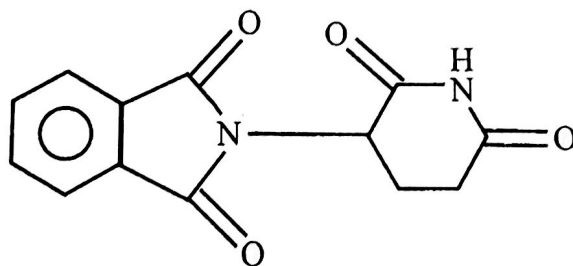


Fig. 7.1

- (i) Explain why proline has a high melting point of 223 °C.
- (ii) Draw the structure of proline in a neutral solution.
- (iii) Write equations to show the buffering action of proline.

[5]

(b) **Fig. 7.2** shows the structure of thalidomide, a sedative drug.



thalidomide

Fig. 7.2

- (i) Draw the structure of the product formed when thalidomide reacts with
 1. $\text{Br}_{2(\text{aq})}$,
 2. acidic solution.
- (ii) Explain why thalidomide exists in two different forms.

6031/3 J2019

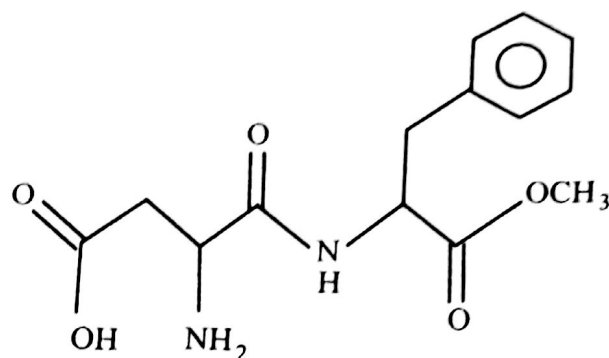
[Turn over

- (iii) Suggest how an unknown sample of thalidomide can be identified using solubility tests.

[10]

[Total: 15]

- 8 (a) Fig. 8.1 shows the structure of aspartame, a methyl ester of the dipeptide of two amino acids.



aspartame

Fig. 8.1

- (i) Write the structures of any **two** functional groups present in aspartame.
- (ii) Comment on the solubility of aspartame in water.
- (iii) Draw the structural formulae of the products for the complete hydrolysis of aspartame in aqueous HCl.
- (iv) Name the **two** amino acids from which aspartame is made.

[8]

- (b) Compare the basic strengths of NH_3 , $\text{CH}_3\text{CH}_2\text{NH}_2$ and $\text{C}_6\text{H}_5\text{NH}_2$.

[3]

- (c) The disposal of condensation polymers in landfill sites is more environmentally acceptable than the disposal of addition polymers.

- (i) Distinguish between a condensation polymer and an addition polymer.
- (ii) Explain why the disposal of condensation polymers in landfill sites is more acceptable.

[4]

[Total: 15]

Section D

Answer any *one* question from this section

9 (a) Explain why

- (i) blocks of magnesium are attached to underground iron water pipes,
- (ii) galvanised corrugated iron roofing sheets do not rust even if the zinc coating is scratched,
- (iii) tin food cans quickly rust when the tin coating gets scratched.

[6]

(b) During preparation, phenylamine, is isolated from the mixture by steam distillation.

- (i) Describe how the steam distillation is carried out.
- (ii) Explain why simple distillation is **not** an appropriate method of isolating the components.

[6]

(c) Fig. 9.1 shows ligand substitution reactions that occur with $\text{Cu}_{(\text{aq})}^{2+}$

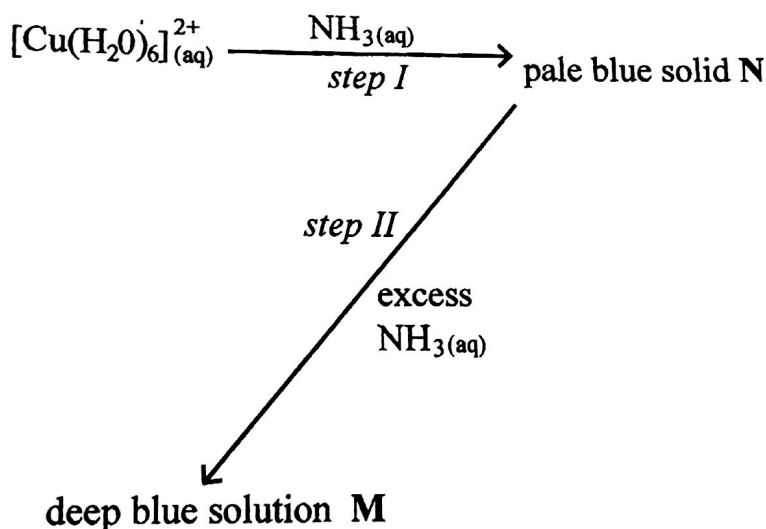


Fig. 9.1

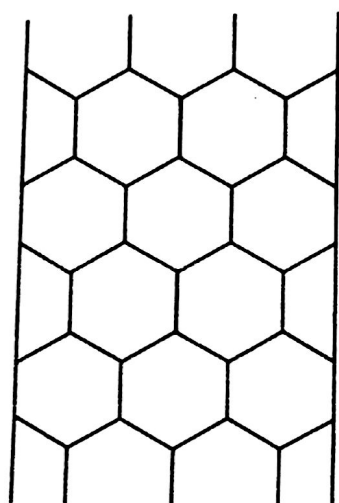
- (i) State the colour of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ aqueous solution.

(ii) Write the formula of

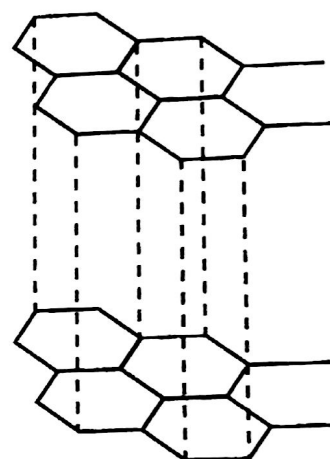
1. solid N,
2. solution M.

[3]
[Total: 15]

- 10 (a) Outline any **two** advantages of sorting waste material before disposal. [2]
- (b) State any **three** problems associated with land filling. [3]
- (c) **Fig. 10.1** shows the structures of two allotropes of carbon, **A** and **B**. The c-c bond length in **A** is about 0.142 nm.



A



B

Fig. 10.1

- (i) Explain why allotrope **A** is classified as a nano material.
- (ii) Name the carbon allotropes **A** and **B**. [5]
- (d) Comment on the potential hazards of using nano materials. [5]
- [Total: 15]